OPTIMAL WINDOWED TEST FUNCTIONS FOR BOUNDING ANALYTIC RANKS OF ELLIPTIC CURVES

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Abstract. Assuming the Generalized Riemann Hypothesis, the explicit formula for the L-function of an elliptic curve $E/\mathbb{Q}$ bounds the analytic rank of E in terms of the traces a_p for $p < e^{2L}$, for any admissible test function supported on a window $[-L, L]$. In the literature this test function is fixed in advance, typically to a Fejér-type kernel. We instead choose, for each individual curve, the test function that minimises the resulting bound within a finite-dimensional space supported on $[-L, L]$. The minimiser is the solution of a linear system with the Gram matrix of the Weil quadratic form of E, and the optimal bound equals the reciprocal of the corresponding reproducing kernel at the central point, which we call the *radix Riemanniana* of E. Because the Fejér-type kernel lies in this space, the optimised bound is never weaker. We certify every bound with ball arithmetic, check the prime-side evaluation against the zero side computed independently from the zeros of $L(E, s)$, and benchmark against the fixed kernel on 1642 curves from Cremona's tables, 2500 curves from the Stein–Watkins database and a set of curves with conductors up to $3 \cdot 10^{16}$. Depending on the sample and on the stopping criterion, the number of primes required to pin down the analytic rank decreases by 8–23% in aggregate. For individual curves of large conductor the decrease ranges from 0% to 43%.

1. Introduction

Let $E/\mathbb{Q}$ be an elliptic curve of conductor N and let $r_{\mathrm{an}}(E)$ be the order of vanishing of $L(E, s)$ at the central point. Under the Generalized Riemann Hypothesis (GRH), Mestre's explicit-formula method [9] bounds $r_{\mathrm{an}}(E)$ from above by a quantity computed from the traces of Frobenius a_p for p below a cutoff. Bober [2] made this practical with a Fejér-type test function, and Klagsbrun, Sherman and Weigandt [7] used the same function to show that the Elkies curve has rank 28 subject only to GRH. In these works the shape of the test function is fixed, and only its support (the cutoff) is varied.

This note asks a simple question: *for a given curve and a given cutoff, which test function gives the best bound?* Within any finite-dimensional space of admissible functions supported on $[-L, L]$ the answer is explicit. The explicit formula expresses the bound as a ratio of a quadratic form (the Weil form of E, evaluated through the a_p) to the square of a linear functional. The minimiser is therefore obtained from one linear solve with a Gram matrix. The optimal value is the reciprocal of the reproducing kernel at 0 of the space equipped with the Weil inner product.

What is known, and what is new. The Weil quadratic form restricted to functions supported on a finite interval, and its Gram matrices with respect to a basis, have been studied for the Riemann zeta function by Yoshida [14], by Connes, Consani and Moscovici [4], by Suzuki [13] via screw functions and de Branges spaces, and recently by Shi [11] as finite "Prime–Weil matrices". Reproducing-kernel methods for Fourier optimisation problems attached to *families* of L-functions, where the input is the limiting one-level density rather than data of an individual L-function, are due to Carneiro, Chirre and Milinovich [3]. To the best of our knowledge, which

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rests on a literature search that is not exhaustive, the optimisation of the test function *for an individual elliptic curve, using its actual a_p*, with certified output and a systematic comparison against the fixed kernel, has not been carried out before. That optimisation is the contribution of this note. The underlying theory is classical. The contribution is the per-curve optimisation, its certification, and the measurements.

Results in brief.

- The optimised bound is never weaker than the fixed-kernel bound at the same cutoff (Proposition 3.4). This is a consequence of the construction, not an empirical finding. The empirical question is *how much* it gains.
- Every reported bound is certified with ball arithmetic. No certified bound in our tests falls below the known rank.
- The prime-side value of the Weil form agrees with the zero-side value computed from the zeros of $L(E,s)$. The discrepancy halves with each doubling of the height up to which zeros are included (Section 5).
- The number of primes needed to determine the analytic rank (Section 6) decreases in aggregate by 8.3–15.2% (Cremona, $N < 4.1 \cdot 10^5$) and by 17.0–22.7% (Stein–Watkins, $4.4 \cdot 10^6 \le N \le 7.5 \cdot 10^9$), depending on the stopping criterion. For individual curves of conductor up to $5 \cdot 10^{15}$ the decrease ranges from 0% to 43%.

2. The explicit formula and rank bounds

Write $L(E,s) = \sum_{n\ge1} a_n n^{-s-1/2}$ so that the central point is $s = 1/2$, and put

$$\Lambda(E,s) = N^{s/2}(2\pi)^{-s}\,\Gamma(s+\tfrac12)\,L(E,s), \qquad \Lambda(E,s) = \pm\Lambda(E,1-s).$$

For $p \nmid N$ let α_p, β_p be defined by $\alpha_p + \beta_p = a_p/\sqrt{p}$, $\alpha_p\beta_p = 1$, and set $c_{p^k} = \alpha_p^k + \beta_p^k$. For $p \mid N$ set $c_{p^k} = (a_p/\sqrt{p})^k$ (so $c_{p^k} = 0$ for additive reduction). Then $-\frac{L'}{L}(E,s) = \sum_{n\ge2} \Lambda(n) c_n n^{-s}$ in the normalisation above.

Let $f\colon \mathbb{R} \to \mathbb{R}$ be even, bounded and supported in $[-L, L]$. Put $g = f * f$, so that g is even and supported in $[-2L, 2L]$, and

$$\hat f(r) = \int_{\mathbb{R}} f(x) e^{irx}\,dx, \qquad h(r) = \hat g(r) = \hat f(r)^2.$$

Writing the nontrivial zeros as $\rho = \frac12 + i\gamma$ (with multiplicity), the explicit formula reads

$$\sum_{\gamma} h(\gamma) \;=\; g(0)\log\frac{N}{4\pi^2} \;+\; 2\Big[-\gamma_{\mathrm E}\,g(0) + \int_0^\infty \frac{g(0)-g(u)}{e^u-1}\,du\Big] \;-\; 2\sum_{n\ge2}\frac{\Lambda(n)\,c_n}{\sqrt{n}}\,g(\log n), \tag{1}$$

where $\gamma_{\mathrm E}$ is Euler's constant. Only prime powers $n < e^{2L}$ contribute. We denote the right-hand side by $W_E(f)$.

Lemma 2.1. *Assume GRH for $L(E,s)$. Then for every nonzero f as above,*

$$r_{\mathrm{an}}(E) \;\le\; \frac{W_E(f)}{\hat f(0)^2}.$$

Proof. Under GRH every γ is real. Since f is real and even, $\hat f$ is real, so $h(\gamma) = \hat f(\gamma)^2 \ge 0$. The central zero contributes $r_{\mathrm{an}}(E)\,\hat f(0)^2$ to the left-hand side of (1), and every other term is nonnegative. □

Remark 2.2. For $f = \mathbf{1}_{[-L,L]}$ one has $g(u) = (2L - |u|)_+$ and $h(r) = 4\sin^2(Lr)/r^2$. Up to scaling, this is the Fejér-type test function used in [2, 7]. We call the resulting bound $B_{\mathrm{std}}(L)$.

3. The optimal window kernel

The map $f \mapsto W_E(f)$ is a quadratic form, $W_E(f) = \mathcal{B}_E(f, f)$, where $\mathcal{B}_E$ is obtained from (1) by replacing $g = f * f$ with the symmetrised correlation of two functions. Fix the basis

$$b_k(x) = P_{2k}(x/L)\,\mathbf{1}_{[-L,L]}(x), \qquad 0 \le k < K,$$

of even polynomials on the window, where P_m is the Legendre polynomial of degree m. Let $G_{ij} = \mathcal{B}_E(b_i, b_j)$ be the Gram matrix and $v_i = \hat{b}_i(0) = \int b_i$, so that $v = (2L, 0, \dots, 0)$. For $f = \sum c_k b_k$ we have $W_E(f) = c^{\mathsf{T}} G c$ and $\hat{f}(0) = v^{\mathsf{T}} c$.

Proposition 3.1. *If G is positive definite, the minimum of $W_E(f)/\hat{f}(0)^2$ over nonzero f in* $\operatorname{span}(b_0, \dots, b_{K-1})$ *equals*

$$B_{\mathrm{rad}}(L) = \frac{1}{v^{\mathsf{T}} G^{-1} v},$$

and is attained at $c^* = G^{-1}v/(v^{\mathsf{T}}G^{-1}v)$.

Proof. Minimise $c^{\mathsf{T}}Gc$ subject to $v^{\mathsf{T}}c = 1$ (Lagrange multipliers, or Cauchy–Schwarz in the inner product $\langle x, y\rangle = x^{\mathsf{T}}Gy$). □

Since $v = \Phi(0)$, the quantity $v^{\mathsf{T}}G^{-1}v$ is the value at the central point of the following kernel.

Definition 3.2 (Radix Riemanniana)**.** Let $\Phi(z) = (\hat{b}_0(z), \dots, \hat{b}_{K-1}(z))^{\mathsf{T}}$. For an elliptic curve $E/\mathbb{Q}$ and a window $[-L, L]$, the *radix Riemanniana* of E (of order K) is the kernel

$$\mathcal{R}_{E,L}(w, z) \;=\; \Phi(w)^*\, G^{-1}\, \Phi(z),$$

that is, the reproducing kernel of the space $\{\hat{f} : f \in \operatorname{span}(b_k)\}$ equipped with the Weil inner product $\mathcal{B}_E$. It is built from the Gamma factor, the conductor and the primes $p < e^{2L}$ alone, without reference to the zeros of $L(E, s)$. The optimal bound of Proposition 3.1 is

$$B_{\mathrm{rad}}(L) = \frac{1}{\mathcal{R}_{E,L}(0, 0)}.$$

Remark 3.3. As $K \to \infty$ the radix Riemanniana is the window analogue, for $L(E, s)$, of the reproducing kernels of the Hilbert spaces of entire functions attached to the Weil form of ζ in [13]. The name and the construction from prime data go back to the author's work on Weil positivity for ζ.

Proposition 3.4. *For every L with G positive definite, $B_{\mathrm{rad}}(L) \le B_{\mathrm{std}}(L)$.*

Proof. $\mathbf{1}_{[-L,L]} = b_0$ lies in the span, so $B_{\mathrm{std}}(L)$ is one of the values minimised over in Proposition 3.1. □

Positive definiteness of G holds under GRH except in degenerate cases, since $W_E(f) = \sum_\gamma \hat{f}(\gamma)^2$. In practice we simply attempt a Cholesky factorisation and discard the window if it fails. Note that *the certificate in Section 4 does not depend on optimality or on positive definiteness*: it evaluates $W_E(f)$ for the specific f that was found.

4. Certification

The optimisation is carried out in double precision. The resulting coefficients are converted to exact rationals $c_k \in \mathbb{Q}$, and L is taken to be an exact decimal. For this fixed $f = \sum c_k b_k$ we evaluate $W_E(f)/\hat{f}(0)^2$ rigorously in ball arithmetic (Arb [1] via `python-flint`, 256-bit precision):

(1) On $[0, 2L]$, $g = f * f$ is a polynomial with rational coefficients. We compute it exactly in $\mathbb{Q}[u]$ and obtain $\hat{f}(0)$ exactly.
(2) The archimedean integral is split at $u = 1/4$. On $[0, 1/4]$ we integrate $(g(0) - g(u))/u$ against the Bernoulli expansion $u/(e^u - 1) = \sum_n B_n u^n/n!$, truncated after 80 terms. The tail is bounded with $|B_n|/n! \le 4(2\pi)^{-n}$. On $[1/4, 2L]$ the integrand is analytic and is integrated rigorously by Arb's certified Gauss–Legendre integration. The constants γ_{E}, π and the logarithms are balls.
(3) The prime sum runs over all prime powers $n < e^{2L}$. The quantities $\log n$, $\sqrt{n}$ and c_n are balls computed from the exact integers a_p. If the ball for $\log n$ contains $2L$, the window is rejected.

The output is a ball containing $W_E(f)/\hat{f}(0)^2$. A statement $r_{\mathrm{an}}(E) < r + 1$ is recorded only if the upper end of the ball is below $r + 1$. The a_p are computed with PARI/GP [10].

5. Validation against the zeros

Identity (1) lets us check the whole pipeline (Rust optimiser → rational coefficients → certified prime side) against an independent computation. We compute the zeros γ of $L(E, s)$ up to height T with PARI's `lfunzeros` and evaluate the zero side $r_{\mathrm{an}}\hat{f}(0)^2 + \sum_{0<|\gamma|\le T} \hat{f}(\gamma)^2$ for the same certified f. Since every omitted term is nonnegative, the truncated zero side must lie below the prime side, and the difference must tend to 0 as $T \to \infty$.

curve	r_{an}	L	prime side (certified)	zero side, $\lvert\gamma\rvert \le T$	T	difference
37a1	1	0.40	1.9967148280	1.9923507969	600	$4.4 \cdot 10^{-3}$
389a1	2	0.65	2.9926821162	2.9903944027	600	$2.3 \cdot 10^{-3}$
5077a1	3	0.90	3.9514652620	3.9497072474	600	$1.8 \cdot 10^{-3}$
234446a1	4	1.15	4.9323755046	4.9292024883	300	$3.2 \cdot 10^{-3}$

Table 1. Certified prime side versus zero side, both normalised by $\hat{f}(0)^2$, for the optimised test function at the cutoff where the bound first drops below $r_{\mathrm{an}} + 1$.

For 37a1 the difference is $8.1 \cdot 10^{-3}$, $4.4 \cdot 10^{-3}$ and $2.4 \cdot 10^{-3}$ at $T = 300$, 600 and 1200 (446, 1024 and 2313 zeros). It roughly halves with each doubling of T, consistent with a truncation tail of order $(\log T)/T$.

6. Benchmarks

For each curve we scan L on a grid of step 0.05 with $K = 10$ basis functions. For both B_{std} and B_{rad} we record the first L at which the certified bound drops below $r_{\mathrm{an}} + 1$ (criterion A), and the first L at which it drops below $r_{\mathrm{an}} + 2$ (criterion B; together with the root number, this already determines the parity of r_{an}). Cost is measured as the number of primes $p < e^{2L}$ used. Because of the grid, prime counts at the threshold are quantised. Ties are therefore frequent and slightly understate the gain.

6.1. **Cremona's tables.** We take one curve per isogeny class, sampled up to 150 per rank from three blocks of Cremona's tables [5] ($61 \le N \le 409989$, 1642 curves, ranks 0–3), with the cutoff $L \le 4.5$ (1028 primes). Table 2 reports the results. In total 6449 bounds were certified, and none lies below the known rank.

	curves decided (std / rad)	rad fewer / equal / more	primes saved
criterion A ($< r_{\mathrm{an}} + 1$)	1573 / 1592	578 / 1064 / 0	15.2%
criterion B ($< r_{\mathrm{an}} + 2$)	1642 / 1642	263 / 1379 / 0	8.3%

criterion A, by rank	0	1	2	3
curves (both decided)	381	450	450	292
primes saved	18.8%	4.3%	3.9%	4.5%

TABLE 2. Cremona sample. "Primes saved" is $1 - \sum P_{\mathrm{rad}} / \sum P_{\mathrm{std}}$ over curves decided by both methods.

6.2. **The Stein–Watkins database.** From the Stein–Watkins database [12] (files of the 2011 distribution) we sample 250 curves per (analytic) rank $0, 1, 2, 3, \geq 4$ from two groups: general conductors $4.4 \cdot 10^6 \leq N \leq 9.7 \cdot 10^7$ and prime conductors $3.6 \cdot 10^9 \leq N \leq 7.5 \cdot 10^9$, giving 2500 curves in total. We check every conductor and root number with PARI. The ranks in the database are Watkins' numerically determined analytic ranks. The cutoff is $L \leq 5.75$ (9592 primes). In total 9105 bounds were certified, and none contradicts the database rank.

	decided (std / rad)	rad fewer / equal / more	primes saved
criterion A	2059 / 2096	1306 / 1194 / 0	17.0%
criterion B	2466 / 2484	889 / 1611 / 0	22.7%

criterion B, by rank	0	1	2	3	≥ 4
curves (both decided)	466	500	500	500	500
primes saved	24.7%	15.1%	5.7%	3.1%	2.7%

TABLE 3. Stein–Watkins sample.

6.3. **Large conductors and high rank.** Table 4 lists every curve of an additional test set, including the cases without gain. Prime counts up to e^{16} were handled by streaming the prime sum.

7. LIMITATIONS

- All bounds assume GRH for $L(E, s)$ and concern the analytic rank. Statements about the algebraic rank additionally use points, the root number and, for $r_{\mathrm{an}} \leq 1$, the theorems of Gross–Zagier [6] and Kolyvagin [8].
- The basis size is fixed at $K = 10$ and the cutoff grid at step 0.05. Larger K can only improve B_{rad}, and a finer grid would turn some ties into gains. We have not optimised either choice.
- The certification is currently performed in Python on top of Arb. The optimiser is written in Rust.
- Our samples cover conductors up to $\sim 3 \cdot 10^{16}$ and ranks up to 9. We have not tested curves of record rank such as the Elkies curve, whose a_p are costly to compute. That is where savings in the number of primes would matter most.

curve	N	rank	criterion	primes std / rad	saved
twist of 11a1, $d = 10009$	1101980891	1	B	329 / 302	8%
twist of 389a1, $d = 10009$	38970051509	1	B	3523 / 3523	0%
twist of 11a1, $d = 100049$	110107826411	0	B	2466 / 2065	16%
twist of 5077a1, $d = 10009$	508614271237	1	B	1220 / 1019	16%
twist of 389a1, $d = 100049$	3893813133989	0*	B	17885 / 13613	24%
twist of 5077a1, $d = 100049$	50819766789877	1	B	1220 / 1220	0%
twist of 389a1, $d = 1000033$	389025674423621	0*	B	102084 / 77395	24%
twist of 5077a1, $d = 1000033$	5077335087528853	1	B	340436 / 195040	43%
twist of 11a1, $d = 1000033$	11000726011979	0	B	undecided by both for $L \le 7.5$	
5187563742a1	5187563742	6	A	11 / 10	9%
$[1, -1, 0, -106384, 13075804]$	249649566346838	8	A	21 / 18	14%
$[1, -1, 0, -135004, 97151644]$	32107342006814614	9	A	32 / 29	9%

TABLE 4. Additional curves. Ranks are lower bounds from PARI's `ellrank` (points). *Here the 2-descent leaves rank 0 or 2 open. The root number $+1$ together with the certified bound $r_{\mathrm{an}} < 2$ gives $r_{\mathrm{an}} = 0$ under GRH, and hence rank 0 by Kolyvagin [8].

Data and code availability

Code (Rust optimiser, certification scripts), the sampled curve lists and all certificates are available at `https://github.com/sectio-aurea-q/radix-rank` (archived on Zenodo, DOI: [to be added]). Running `./reproduce.sh quick` re-derives and re-certifies the bounds for seven reference curves in about one minute.

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